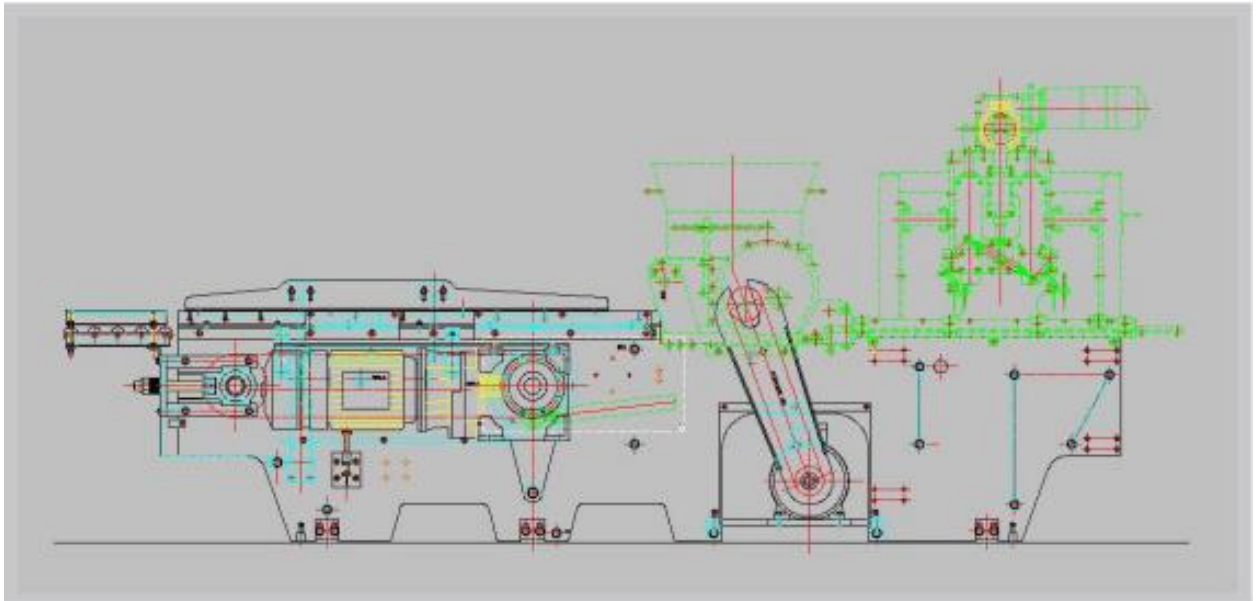


ONUR KIREMIT

MANUFACTURING EQUIPMENT OF ROOF TILES



MAY 2023

CONCRETE CONVEYOR TO THE EXTRUDER BOX

The mixture exit of the hopper must be adjusted, so that it works with more or less quantity at the entrance of the hopper of the extruder, depending on the speed of the production line.

It is necessary to balance the amount of mixture to the belt that feeds the extruder to avoid the constant operation of the feeding belt, since in this way it is not possible to maintain a constant level in the extruder box.

For a better performance of this conveyor, it should be slow in a normal and constant feed to always maintain the same feed column and also have a fast speed when there is no mixture in the conveyor to avoid production stops in the factory due to lack of mixture in the extruder box.



It must be guaranteed that the entrance of concrete into the hopper of the extruder incurs its majority in the area of consumption of the same since otherwise vaults will be formed and the concrete will not flow in the feeding zone of the corresponding rollers and slippers.

Clear example of centering the entrance of mortar in the area of consumption.



Place a metal detector to prevent the entry of metal parts in the extruder conveyor.
(Improvement) (Future investment if TC Roller and slipper will be used)



The scraper does not work well is fixed (deterioration of the rubber band and mixture fall to the ground, it is necessary that the scraper is mobile and maintain the pressure for cleaning the mortar band.



To clean the band, a scraper with "tensioners" in the center of the drum and perpendicular is better.

TENSIONERS :



MANUFACTURING EQUIPMENT OF ROOF TILES

This machine, being the most complex as a whole, has been distributed in the following zones or equipment:

***Pallets Pushing System, Extrusion Head, Bench & Removable Bench**

Guides.(BENCH GUIDES)

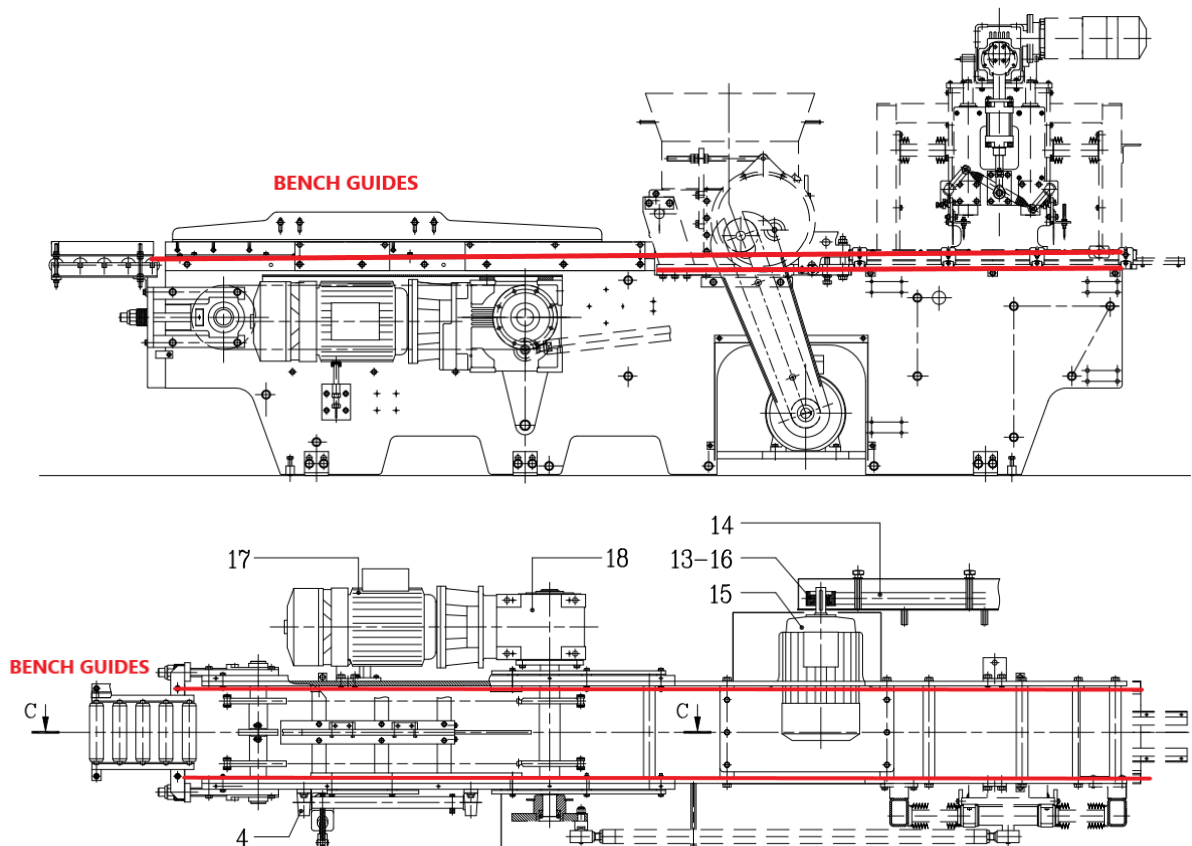
***Pallets Pushing System.**

***Extrusion head “Double Roman”**

***Electromechanical Cutter “Double Roman”**

PALLETS PUSHING SYSTEM, EXTRUSION HEAD, BENCH & REMOVABLE BENCH GUIDES.

It is important to maintain in good condition the bench guides, both the lower (all horizontal and aligned guides) and the lateral ones (all vertical and aligned guide), of the 3 sets where they are located, the pallets aluminium must pass through the guides of these 3 positions well aligned and favouring the entry in the passages of the pushing system to the extrusion head and in turn to the cutter, any difference in the guides can cause a deformation or wear to the moulds in their passage.



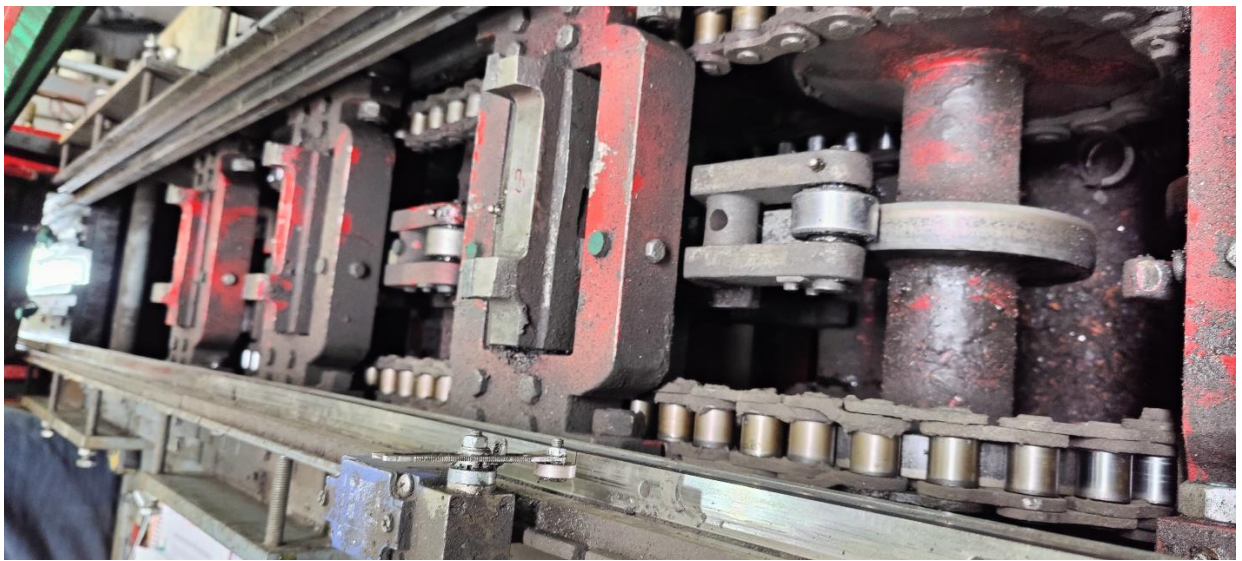
All guides on the bench guides must be kept in good condition and replaced when necessary.



PALLETS PUSHING SYSTEM.

The pushing system chain is not bad in appearance, the trolley polycorers need to be adapted, but the measurements between the pusher fingers are very bad, the initial measurement of this distance is 420 mm, currently most of them have 424 mm between the pusher fingers, the chain and couplings in use have generated a stretching of the chain of about 4 mm, so you can identify a gap of “4 mm” between pallets when pushing the chain.

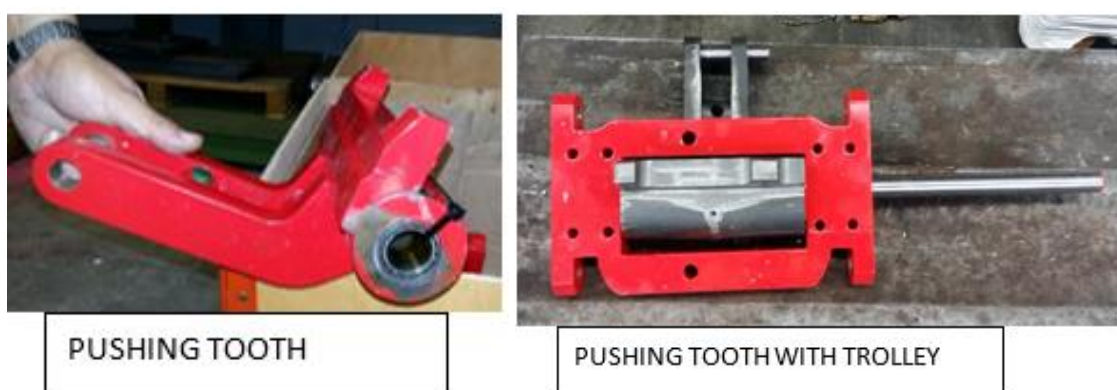
It is recommended to replace the pusher chain if there are synchronisation problems with the “cutting of the tile” or if the distance between its fingers has reached 425 mm, at the same time replace all the bearings and identify if the drive sprocket is worn or not, in case it is necessary to replace it at the same time as the chain.



Replace trolley bearings.



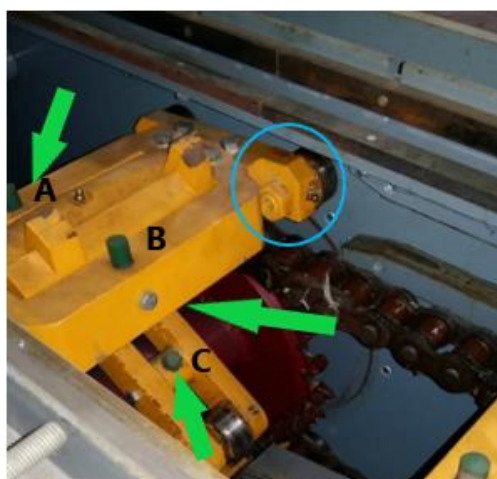
Replace pushing tooth bearings.



If worn or loose, replace the chain drive sprocket.



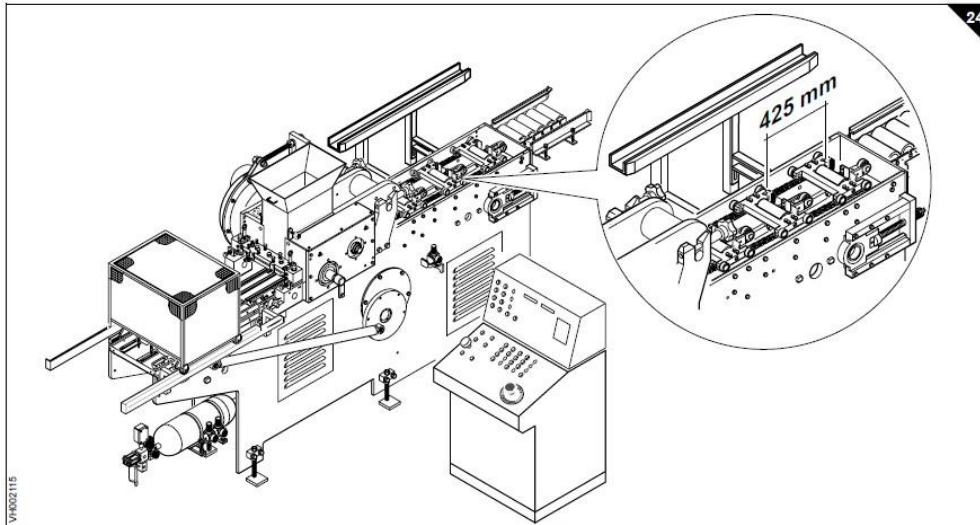
Pushing System Trolleys Polycors (Measurements of polycors)



A	9 mm
B	8 mm
C	8 mm

8.2.12 Checking for wear on the chain system

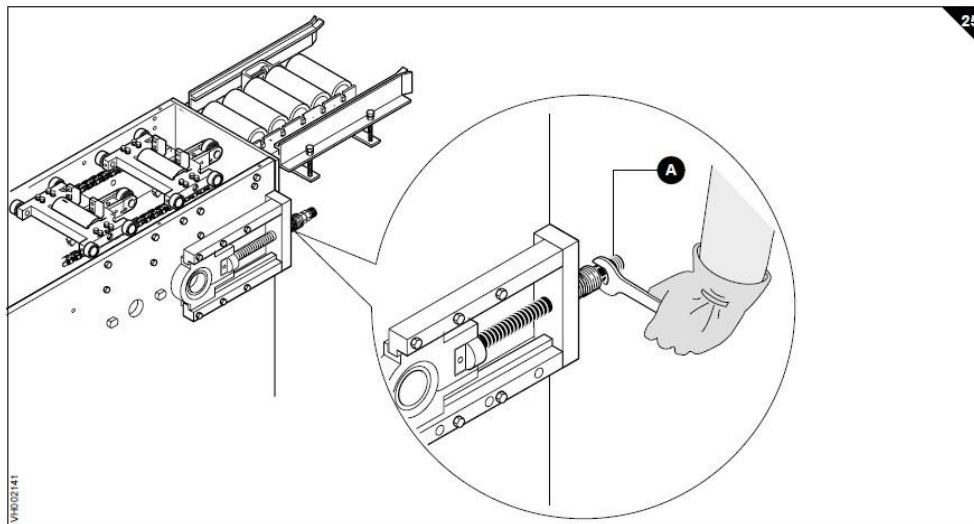
Check the distance between the two pushing teeth on the trolleys at periodic intervals; if the distance is greater than 425 mm, the chains are worn and must be replaced immediately (see fig. 24).



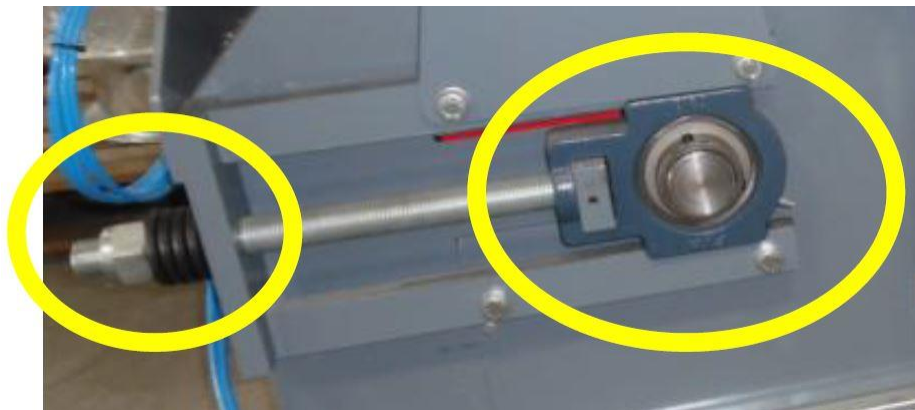
8.2.13 Tensioning the chain system

Check the chain system tension at periodic intervals; the chain needs tensioning if the pushing unit starts getting noisier and the links are too slack.

Tighten each of the adjustment screws the same amount (A) (see fig. 25)



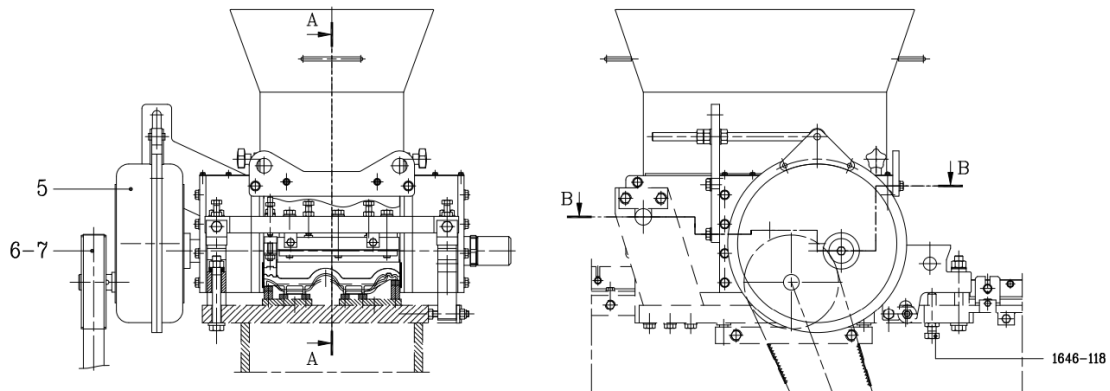
Check the chain tension measurements periodically, on both sides equally.



EXTRUSION HEAD

This is the main equipment in the manufacture of concrete roof tiles and where the right settings are needed to achieve a quality tile with the mix produced by the mixer.

In the event of a mix with errors in the additions of raw materials or problems with moisture or problems with the height adjustments of the rollers and slippers, this equipment can suffer high compressions, pressures, tensions and wear between its elements (ROLLERS, SLIPPERS, STIRRERS, SHAFTS, BEARINGS, SPROCKETS, CHAINS) when producing concrete tiles



The 2 problems with the Rollers in the extruder and the space that occurs in the assembly of the Big and Small slipper in the slipper support are discussed in a special ANNEX.



EXTRUSION HEAD PROCESS SHEET

Starting from the premise that we are going to manufacture concrete roof tiles in the extruder with an initial mix of pre-set characteristics (sand, cement, pigment, water, additive and with a humidity that we intend to be stable), it is necessary to establish some values for adjusting the rollers/slippers, the speeds of the Rollers and Stirrers and the rest of the extrusion values, which we indicate in our Process Sheet, to serve as a basis for identifying what is generated by any change in the variables that we can modify with the aim of improving the Quality of the Tile and always producing within the control values of the Process Sheet.

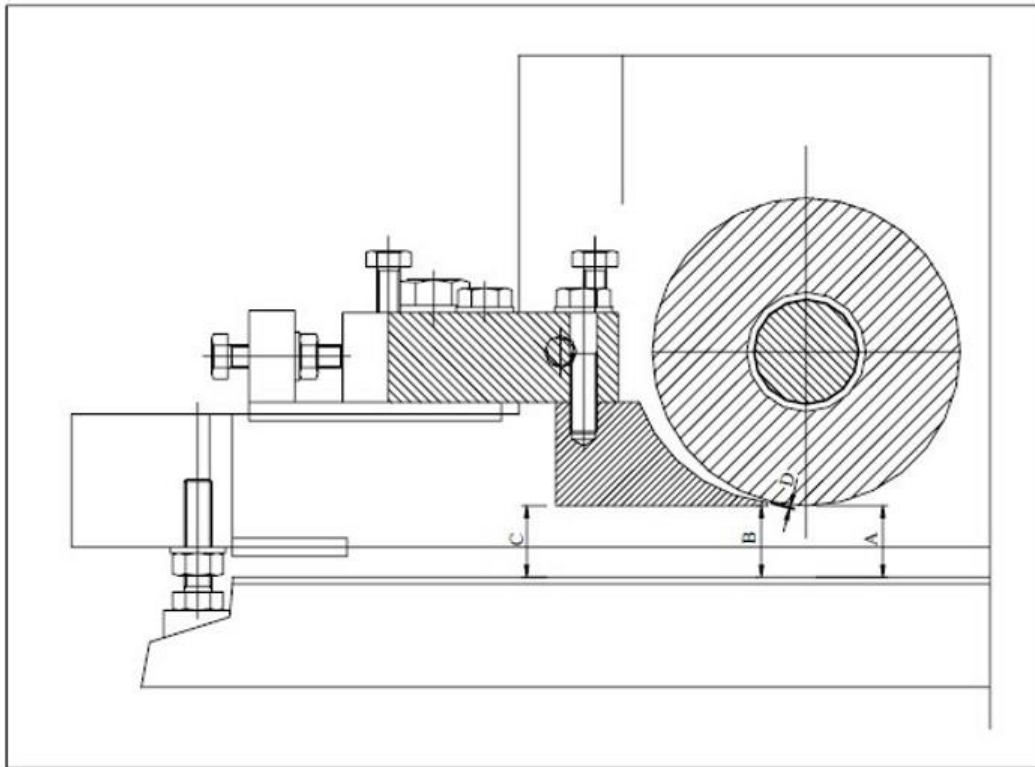
PROCESS SHEET							Code
							2023
							Review number:
							Page: 1
FACTORY: ONUR							
SECCION : EXTRUDER BOX							
PROCESS : HEAD EXTRUSION / DOUBLE ROMAN							
PROFILE	Nº	PARAMETERS	NOMINAL VALUE	MINIMUM VALUE	MAXIMUM VALUE	UNITS	NOTES
STEEL EXTRUDER BOX	1	Roller speed.				r.p.m.	
	2	Star bar speed.				r.p.m.	
	3	Roller height.(A)				mm	
	4	Slipper entry height (B)				mm	
	5	Slipper exit height.(C)				mm	
	6	Big Roller to Big Slipper "GAP" (D)				mm	
	7	Small Slipper inclination (+ 2 mm)				mm	
	8	Limiters length . (policord)				mm	
	9	New roller diameter.				mm	
	10	Used roller diameter.				mm	
	11	Height distance between big and small slipper				mm	
	12	Maximum roller temperature				°C	
	13	Maximum slipper temperature				°C	
	14	Roller ammeter				Ampere	
	15	Pushing chain ammeter				Ampere	
	16	Column height.				mm	
	17	Tiles per minute				t.p.m	



The EXTRUDER PROCESS SHEET that I am going to present consists of 17 indicators that I consider "necessary to take into account", in order to control the whole "Extruder PROCESS SHEET". (After the implementation of the PROCESS SHEET and the monitoring of its indicators, it can be assessed whether any of the 17 indicators do not need to be monitored.)

In the development of this PROCESS, each factory can expand or reduce the proposed indicators as they see fit.

There are 4 main indicators in this Process, which are the measures of adjustment of the tile head, **A**, **B**, **C** and **D**, they are very particular for each model of tile and conditioned to **the raw materials (mixture)** that are used for the manufacture of the tile.



The measurements **"A"**, **"B"**, **"C"** and **"D"** correspond to:

"A" = Distance from the bench guides to the flat areas of the roller, according to the thickness of the tile in the flat area minus the height of the mould.

"B" = Distance from the bench guides to the Big Slipper entrance.

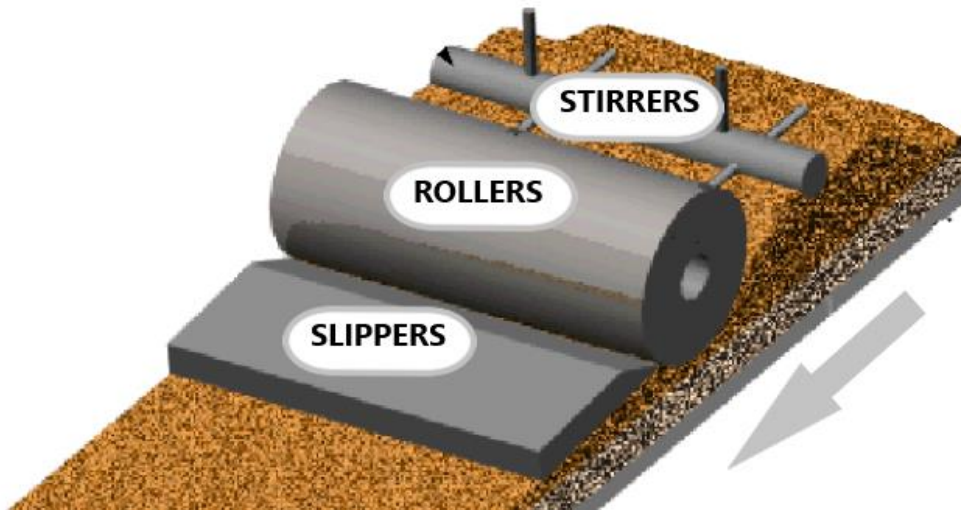
"C" = Distance from the bench guides to the exit of the Big Slipper.

("Track slipper inclination mm" = **B - C**)

"D" = Distance from the Big slipper to the Big Roller **"GAP D"**.

Initially for measures **B**, **C** and **D**, there are reference values, as detailed in the table below, always depending on the type of sand (fine / coarse) used.

SLIPPER REFERENCE DISTANCES			
Ref.	Distance	Type of sand	Initial value
A	Distance of roller + pallet	/	10 mm
B	$C + 0.8 \div 1 \text{ mm}$	With fine sand	/
B	$C + 1 \div 1.3 \text{ mm}$	With coarse sand	/
C	$A + 0.8 \div 1 \text{ mm}$	With fine sand	/
C	$A + 1.1 \div 1.3 \text{ mm}$	With coarse sand	/
D	$2 \div 3 \text{ mm}$	/	/



There is also an initial reference roller speed, the calculation values are the speed (tiles/minute) and the smallest roller diameter of the tile profile.

However, once in production with the other settings, it can vary, any excess of the roller speed will result in increased wear of the roller

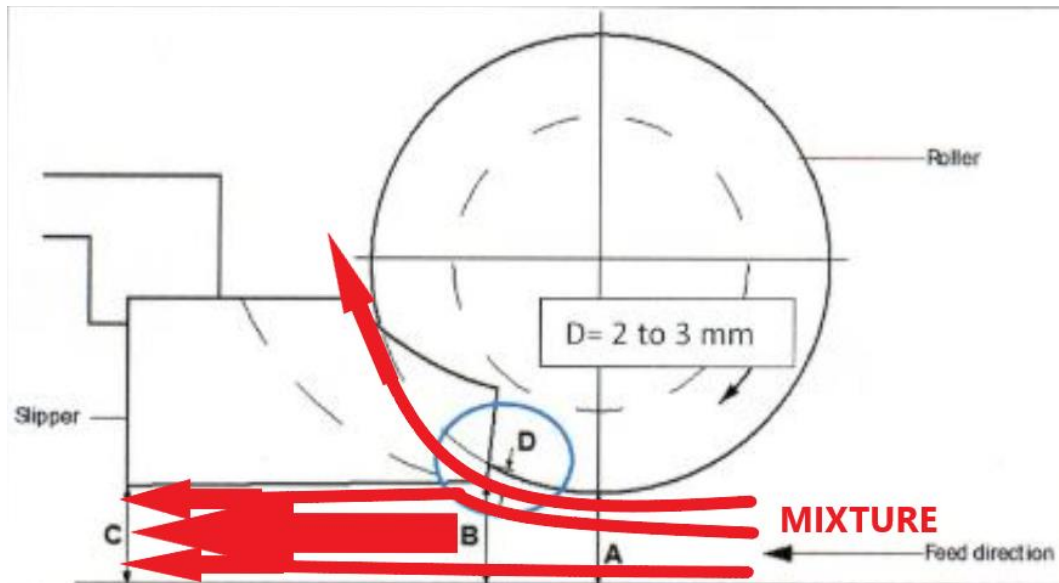
$$\text{Roller Speed (Rpm)} = \frac{(N. \text{ Tiles/Min}) * 42 * 1.2}{(D \text{ min. Roller in cm}) * 3.14} = R.P.M.$$

The 17 indicators presented in this Process Sheet are possible or indicative values, each of which must be confirmed during the production of roofing tiles to validate that they meet the objectives set.

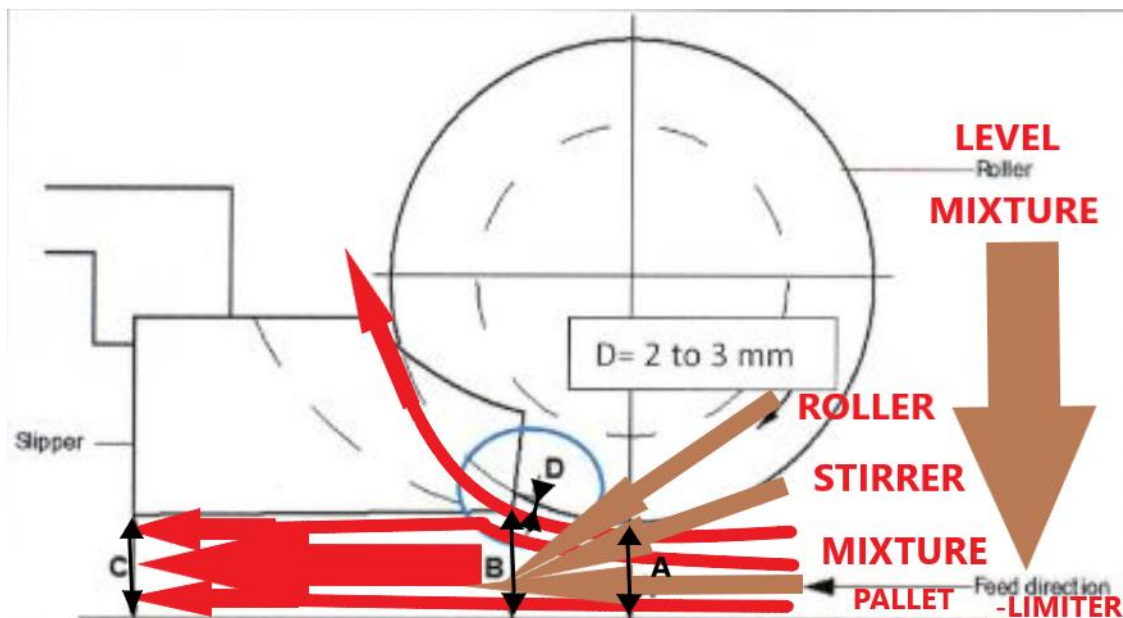
PROCESS SHEET							
							Code
							11/05/2023
							Review number:
							Page: 1
FACTORY: ONUR							
SECCION : EXTRUDER BOX							
PROCESS : HEAD EXTRUSION / DOUBLE ROMAN							
PROFILE	N°	PARAMETERS	NOMINAL VALUE	MINIMUM VALUE	MAXIMUM VALUE	UNITS	NOTES
STEEL EXTRUDER BOX	1	Roller speed.	120	110	130	r.p.m.	CHECK
	2	Star bar speed.	140	130	150	r.p.m.	CHECK
	3	Roller height.(A)	36,6	31	35	mm	CHECK
	4	Slipper entry height (B)	37,6	32	36	mm	CHECK
	5	Slipper exit height.(C)	39,6	33	38	mm	CHECK
	6	Big Roller to Big Slipper "GAP" (D)	2,5	2	3	mm	CHECK
	7	Small Slipper inclination (+ 2 mm)	2	1,5	2,5	mm	CHECK
	8	Limiters length . (policordi)	120	100	150	mm	CHECK
	9	New roller diameter.	134	133,8	134	mm	CHECK
	10	Used roller diameter.	127	126	128	mm	CHECK
	11	Height distance between big and small slipper	3,5	3,0	4,0	mm	CHECK
	12	Maximum roller temperature	35	20	50	°C	CHECK
	13	Maximum slipper temperature	30	20	35	°C	CHECK
	14	Roller ammeter	16,5	15	23	Ampere	CHECK
	15	Pushing chain ammeter	11,5	10	17	Ampere	CHECK
	16	Column height.	500	450	600	mm	CHECK
	17	Tiles per minute	50	48	52	t.p.m	

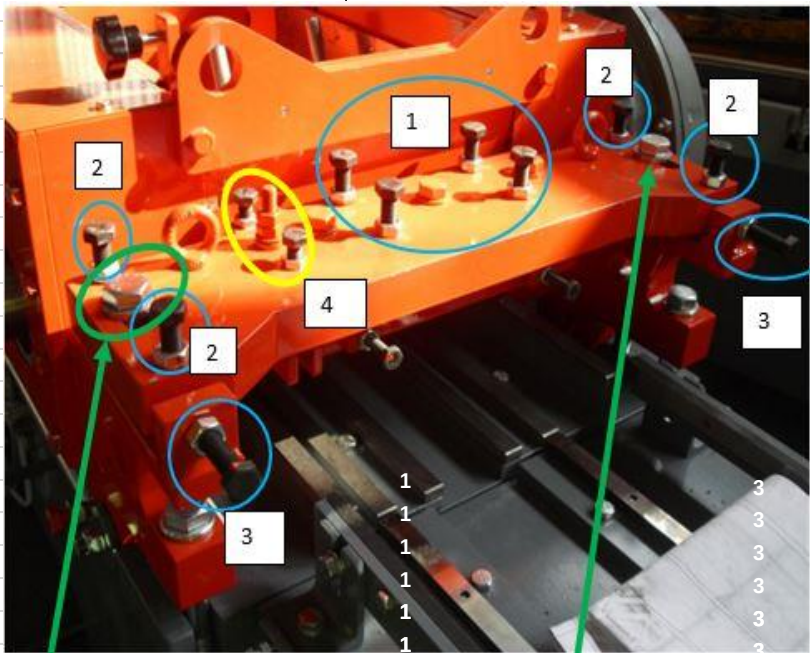
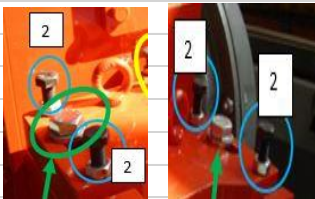
To understand a little more about the shaping of the tile on a mould, the philosophy is to overfeed concrete into the space between the mould and the roller / slipper, to maintain a fresh concrete on the pallet at a certain compaction, so that it does not deform during the whole process of cutting, moving and stacking.

All the mixture from the hopper supported on the base of the pallet, is fed by the pallet itself towards the roller assisted by the rotation of the roller, in that direction with an initial leakage of concrete between the roller and slipper.

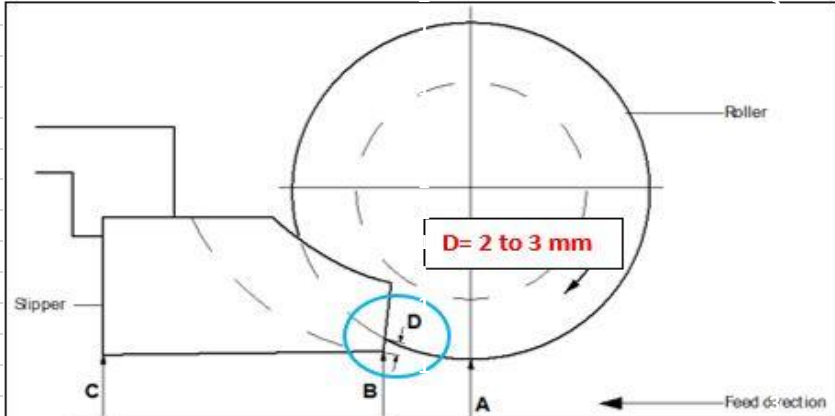
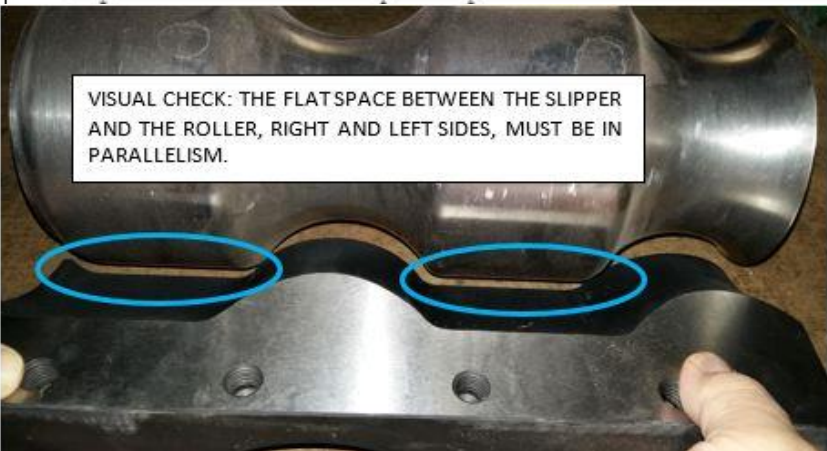
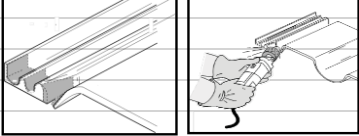


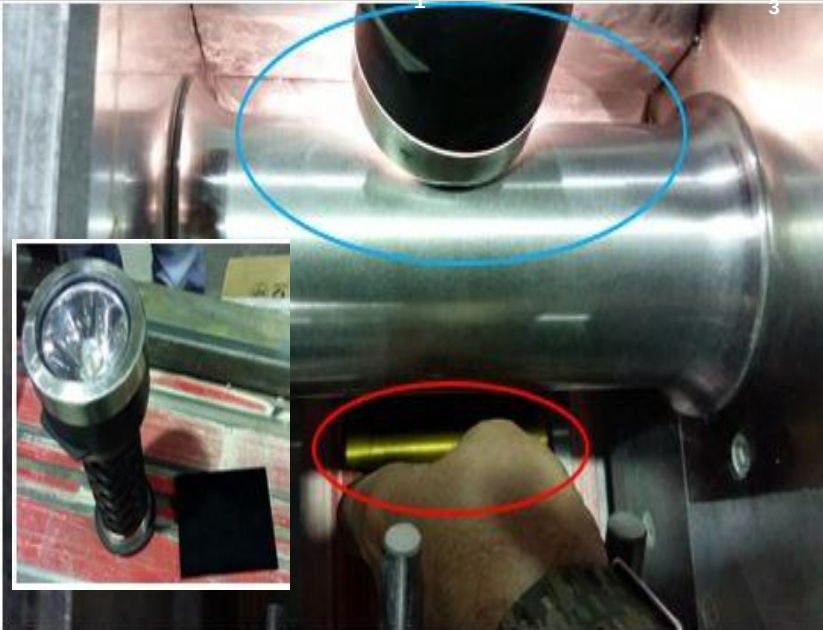
At this point there are already more process indicators, which intervene for this shaping, the height of the mixture column in the hopper (mm), the speed of the roller (rpm), units / dimensions of the stirrers and their speed (rpm), mixture limiters in the flat areas of the tile (YES / NO) (length mm).

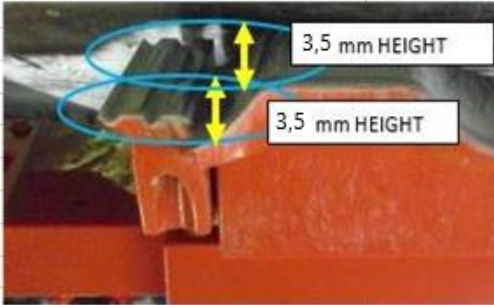


SETTING OF SLIPPER AND SMALL SLIPPER		
the adjusment bolts / nuts		
ACTIONS / NOTES	PLAN	ACTIONS / NOTES
		
SLIPPER UNIT RETAINER BOLTS (LEFT AND RIGHT). SEE SOP 140 REMOVE AND CLEAN THE SLIPPER UNIT.		 N.4 Central adjustment bolts for inclination. (Big Slipper)
		 N.2 (LEFT)+ N.2 (RIGHT) adjustment bolts for HIGH / LOW setting. (Big and small) Slippers.
		 N.1 (LEFT)+ n.1 (RIGHT) front adjustment bolts for horizontal setting, to close the gap between the slipper and the roller: gap "D" (See SOP 260 and sop 270). Unscrew for spacing from roller, screw to come near.
1 N.4 CENTRAL ADJUSTMENT BOLTS FOR INCLINATION.		
2 N.2 (LEFT)+ N.2 (RIGHT) ADJUSTMENT BOLTS FOR HIGH/LOW SETTING.		
3 N.1 (LEFT)+ N.1 (RIGHT) FRONT ADJUSTMENT BOLTS FOR HORIZONTAL SETTING, TO CLOSE THE GAP BETWEEN THE SLIPPER AND THE ROLLER: GAP "D", SEE SOP 260 AND SOP 270. UNSCREW FOR SPACING FROM ROLLER, SCREW TO COME NEAR.		
4 SMALL SLIPPER (SEE YELLOW BOLD CIRCLE): N.2 ADJUSTMENT BOLTS FOR INCLINATION AND HIGH/LOW SETTING. N.1 NUT FOR HORIZONTAL SETTING TO CLOSE THE GAP BETWEEN THE SMALL SLIPPER AND THE SMALL ROLLER.		 Small slipper (See Yellow bold circle): N.2 Adjustment bolts for INCLINATION and HIGH / LOW setting. N.1 nut for horizontal setting to close the gap between the Small Slipper and the Small Roller.

SETTING OF SLIPPER : THE GAP "D"

ACTIONS / NOTES	PLAN	ACTIONS / NOTES
		1) The gap "D" allows the small recirculation of the concrete, reducing the pressure in the extrusion slipper and it determines the right compaction of the concrete to produce the roof tiles. 2) It is important to check visually the measure of the gap "D". 3) The visual check has to be made in the flat space between the slipper and the roller, right and left sides.
		(See image Blue circle). 4) To visual check it is necessary a mirror with a light: (See next SOP 280). 5) The measure of the flat space between the slipper and the roller vary, according to the mix wet or dry, from 1 to 5 mm.
		6) To settle the gap "D" unscrew/ screw the N.1(LEFT)+ N.1(RIGHT) front adjustment bolts of the slipper unit. Unscrew for spacing from roller, screw to come near. (See SOP 250).
		7) The small recirculation of the concrete, taking out the pressure in the extrusion slipper is also obtained with the grinding of the slipper and of the small slipper and with the polycord rope diameter of 20 mm positioned inside the extrusion head.
Q THE QUALITY CONTROL ON THE ROOF TILES DETERMINE THE SETTING OF THE GAP "D".		(See SOP 280 and 290).

CHECK AND SET THE GAP "D"		
ACTIONS / NOTES	PLAN	ACTIONS / NOTES
	To check the measure of the GAP "D":	
		1) Light with a torch the roller and slipper from the top. (See image Blue circle). 2) Put the welding mirror under the roller. Be careful, the space is narrow. (See image Red circle). 3) The welding mirror reflect Gap "D". (See image Red circle).
		4) It is important to verify that the light reflected of the Gap "D" has the same thickness because the flat space between the slipper and the roller, right and left sides, must be in PARALLELISM. The check is visual. (See SOP 260).
		5) To be sure to set in the same way right and left sides, unscrew/ screw the N.1 (RIGHT) front adjustment bolt of the slipper unit, measure the horizontal setted displacement with a vernier (See image with the Yellow arrow) and do the same for the N.1 (LEFT) front adjustment bolt.
		★ 6) Be careful, the screw of the (right/left) front adjustment bolt of the slipper unit, to come near the roller with the slipper, is suggested to be done with setting of 0,5 mm each time !
	Q THE QUALITY CONTROL ON THE ROOF TILES DETERMINE THE SETTING OF THE GAP "D".	The contact of the slipper with the roller can damage the slipper.

SETTING OF SLIPPER WITH THE SLIPPER AND GAP WITH SMALL ROLLER		
ACTIONS / NOTES	PLAN	ACTIONS / NOTES
 3,5 mm HEIGHT 3,5 mm HEIGHT	THE STANDARD SETTING OF THE HEIGHT OF THE SMALL SLIPPER COMPARED TO THE SLIPPER IS : 3,5 mm THE SAME MEASURE IN THE FRONT AND REAR EDGES IS THE CHECK OF THE RIGHT INCLINATION WITH THE SLIPPER.	5) The standard setting of the height of the small slipper compared to the slipper is for Double Roman 3.5 mm;
		6) With a vernier verify the same height of the small slipper compared to the slipper in its front and rear edges, see image, to check the right inclination with the slipper.
	HEIGHT BETWEEN THE ROLLER AND THE SMALL ROLLER.	7) Another way to start the setting is to measure the height between the roller and the small roller. (See image Red circle). This can be the measure to start the setting of the small slipper height compared to the slipper. 8) In any case, verify if the profile of the extruded roof tile is correct or not, and operate consequently

This video from VORTEX has been provided, where all the phases of the Roller and Slipper adjustment are identified.



In addition, it is necessary to monitor the thickness and weights of the wet and dry tiles, the controls should initially be "every hour" (after the implementation of the thickness control, the appropriate frequency of monitoring should be assessed), as there are many fluctuations in the mixture due to the sand or water in the mixture, in the case of the dry tiles it can be started every 2 hours and then, depending on the results, adjusted to the best needs.

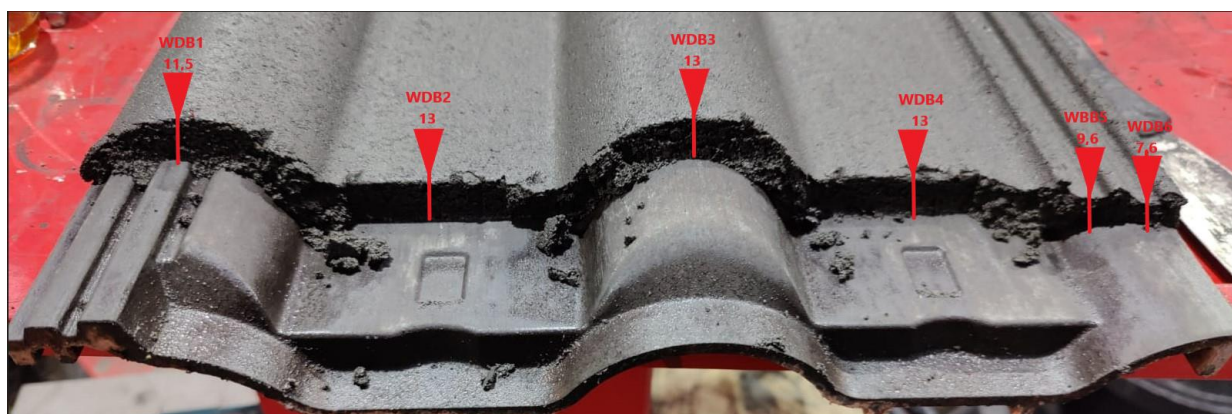
WET AND DRY TILES.

I have varied my initial approach, leaving 6 points of control in the wet tile and another 6 points in the dry tile, the ranges of measures are the initial ones, after the verifications it will be necessary to determine when it is superior to the maximum value the actions to carry out in the extruder.

WET DOUBLE ROMAN TILE MEASURES :

Regarding the control of the measurements of the wet tile, I consider that they can be considered according to the points indicated in the photo.

The suggested thicknesses can be WDR1 to WDR6 at 65mm from the end of the tile.



<u>DOUBLE ROMAN WET TILE MEASURES</u>								
	<u>WDR1</u>	<u>WDR2</u>	<u>WDR3</u>	<u>WDR4</u>	<u>WDR5</u>	<u>WDR6</u>		
	<u>11,5</u>	<u>13</u>	<u>13</u>	<u>13</u>	<u>9,6</u>	<u>7,6</u>	<u>INITIAL</u>	<u>VORTEX</u>
	<u>12,5</u>	<u>11,69</u>	<u>11,2</u>	<u>11,39</u>	<u>11,22</u>	<u>7,9</u>	<u>Mr.Murat</u>	<u>CHECK</u>
<u>RANGE</u>	<u>11....12</u>	<u>2,5....13,</u>	<u>12,5....13,5</u>	<u>12,5....13,5</u>	<u>9,1....10,1</u>	<u>7....8,1</u>		
<u>DATE: DD-MM-YY</u>							<u>4,7....4,6 Kg.</u>	
<u>TIME</u>							<u>WEIGHT</u>	
	<u>12,50</u>	<u>10,75</u>	<u>11,20</u>	<u>11,10</u>	<u>11,8</u>	<u>8,00</u>	<u>4,44</u>	<u>Mr.Murat</u>
	<u>12,52</u>	<u>11,69</u>	<u>11,18</u>	<u>11,39</u>	<u>11,22</u>	<u>7,90</u>	<u>4,70</u>	<u>CHECK</u>
							<u>4,32</u>	
							<u>4,50</u>	

These measurements should be checked again, as they are quite different from the initial theoretical measurements, it may be an error in the reading on the caliper, due to the mixture with the accelerator or the measurements of the slipper.

WET WEIGHT TILE

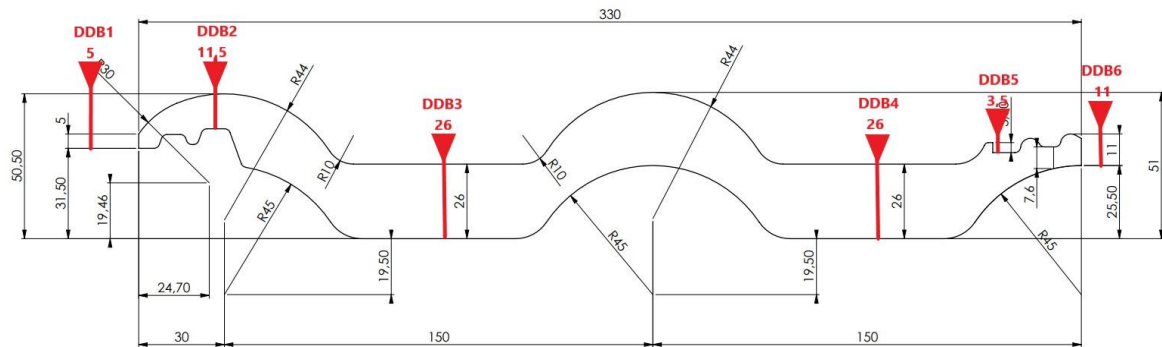
Factory-tested tile weights from **4.28 kg to 4.7 kg** are due to the use of accelerator in the mixture, with these differences in weights it is difficult to determine which weight should be considered as the target weight in the Process Sheet.

In a mixture without accelerator, the theoretical wet weight of the tile, on the curing weight loss (150 g/200 g) , initially **4.7 kg to 4.6 kg** to achieve a tile weight after curing of **4.4 kg to 4.5 kg**.

DRY DOUBLE ROMAN MEASURES :

Suggested thicknesses can be **DDB1 to DDB6 at 65mm** from the end of the tile.

The measurement ranges are the initial ones, after the verifications it will be necessary to determine when it is superior to the maximum value the actions to be carried out in the extruder on the **wet tile**.



DOUBLE ROMAN DRY TILE MEASURES

	DDR1	DDR2	DDR3	DDR4	DDR5	DDR6		
DRAWING	5	11,5	26	26	3,5	11	INITIAL	VORTEX
							Mr.Murat	CHECK
RANGE	5...6	11....12	26....27	26....27	3,5...3	11.....11,5		
DATE: DD-MM-YY							4,4....4,5	Mr.Murat
TIME							WEIGHT	CHECK

I don't have measurements of the dry tile, you have to make them and check what differences there are and what are the objective values for the Process Sheet.

ROLLER AND PALLET PUSHING AMPERAGE CONTROL

I don't have amperage measurements of the pusher chain and roller, you have to identify them on the control screen and check what the differences are and what the target values are for the Process Sheet.

The theoretical amperage consumption in the thrust train under normal circumstances should be **16,5 A** (Pallet Pushing System) and **11.5 A** for the Roller.

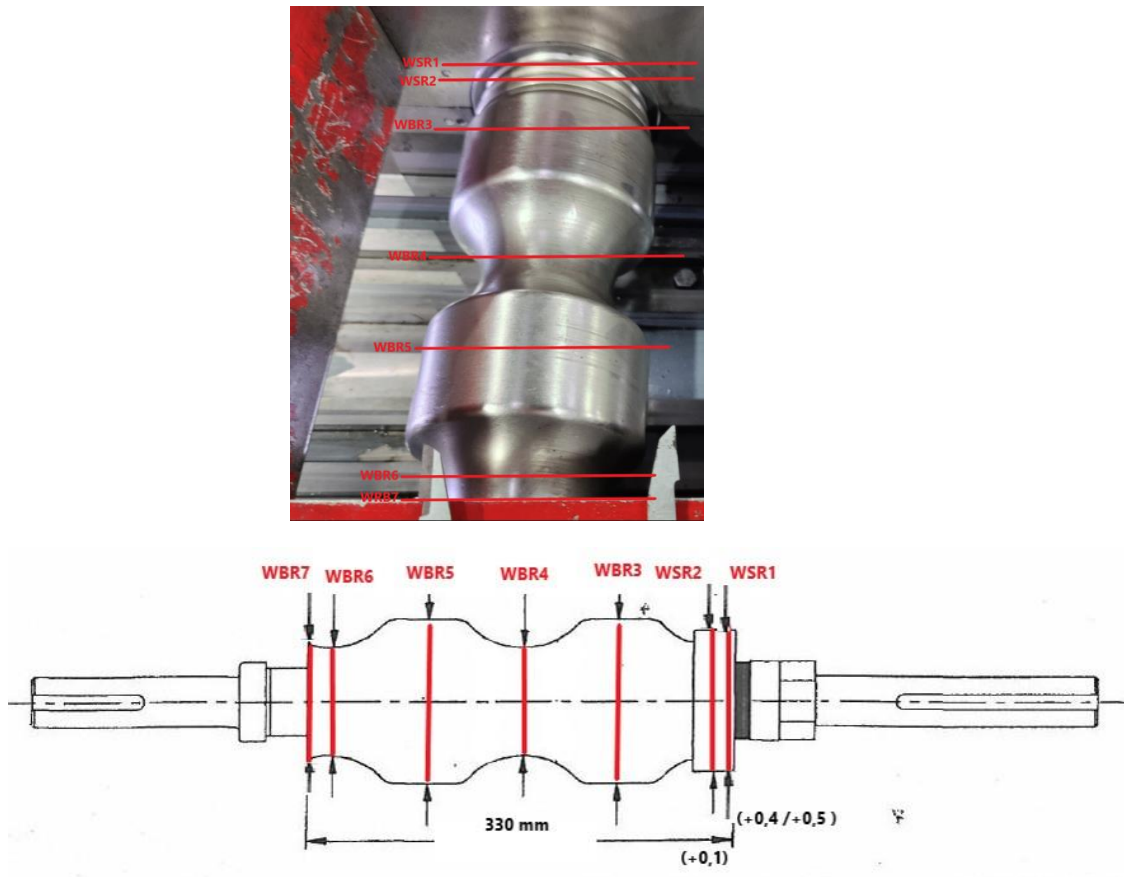
ROLLER AND SLIPPER WEAR CONTROL.

In relation to the established rule on wear, for 1 Big Roller, 2 Small Rollers, 2 Big Slipper and 4 Small Slippers are consumed.

It is necessary to carry out an initial control, every 2 days, depending on the production carried out. Progressively, if it is of interest, a longer period of time between controls can be considered, depending on the wear control.

BIG / SMALL ROLLER DOUBLE ROMAN WEAR MEASURES :

Planned control points **WSR 1 to WBR7**, starting with the **Small Roller** (interlocking) and ending with the **Big Roller** (overlocking).



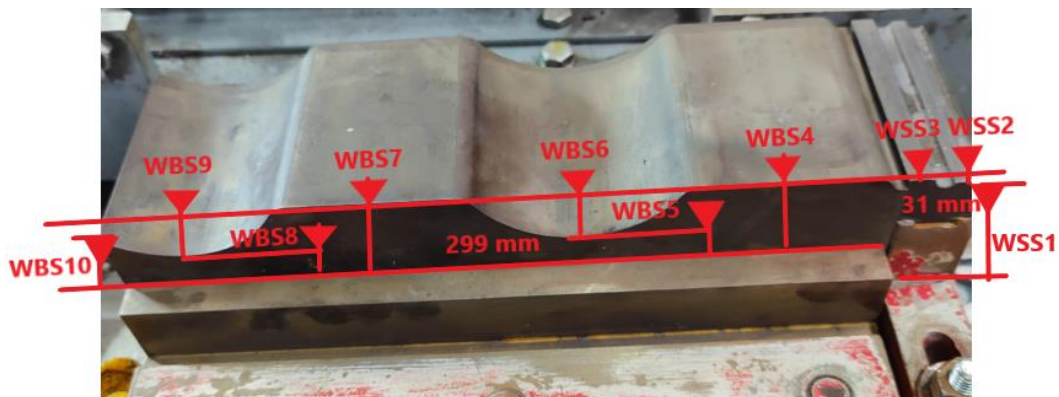
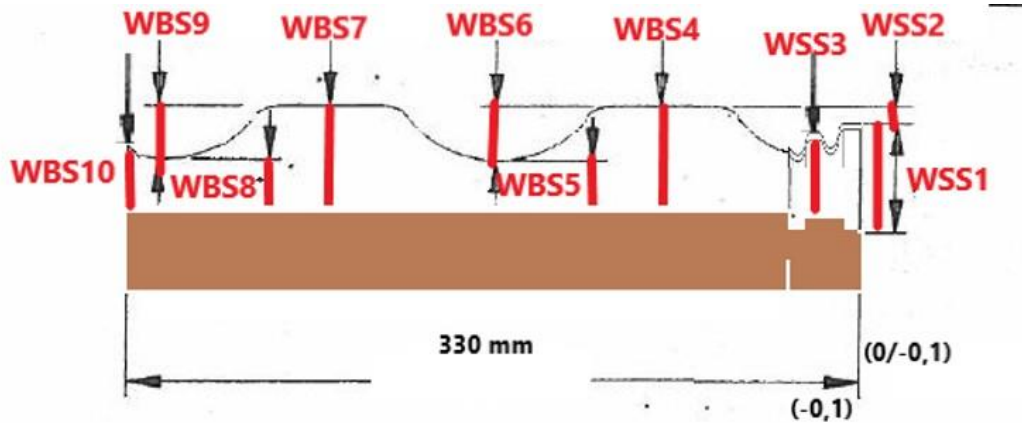
BIG / SMALL ROLLER DOUBLE ROMAN WEAR MEASURES									
	WSR1	WSR2	WBR3	WBR4	WBR5	WBR6	WBR7		
ROLLERS NEW	122	126	134	84	134	85	114	INITIAL	VORTEX
11/05/2023	120,2	126	130,3	80,9	130,3	85	106	Mr.Murat	CHECK
RANGE			134...126		134...126				
DATE								TIME	

These measurements should be checked again, as they are very different from the initial theoretical measurements of the Vortex /Big / Small Roller), it may be an error in the gauge reading or the manufacture of the Rollers to different measurements to the original Vortex ones.

BIG / SMALL SLIPPER DOUBLE ROMAN WEAR MEASURES :

Planned control points **WSS 1 to WBS 10**, starting with the **Small Slipper** (interlocking) and ending with the **Big Slipper** (overlocking).

It is necessary to check all the initial measurements, because the slippers are fixed to slipper supports, which are in most cases the reference points of measurement.



<u>BIG / SMALL LIPPER DOUBLE ROMAN WEAR MEASURES</u>											
	<u>WSS1</u>	<u>WSS2</u>	<u>WBS3</u>	<u>WBS4</u>	<u>WBS5</u>	<u>WBS6</u>	<u>WBS7</u>	<u>WBS8</u>	<u>WBS9</u>	<u>WBS10</u>	<u>INITIAL</u>
SLIPPERS NEW	53	6	4	40	15.5	24.5	40	14.5	25.5	29.5	VORTEX
11/05/2023	52.26	5.56	3.03	40	0	32.6	40	0	24.9	22.1	Mr.Murat
RANGE											CHECK
DATE											TIME

These measurements should be checked again, as they are very different from the initial theoretical measurements of the Vortex /Big / Small Slipper), it may be an error in the gauge reading or the manufacture of the Slippers to different measurements to the original Vortex ones.

STIRRERS and LIMITERS

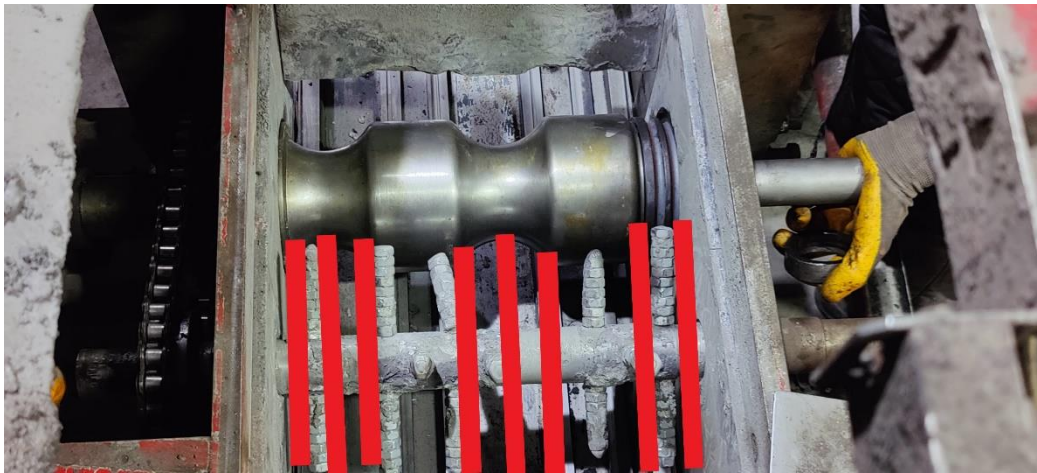
STIRRER PIN FUNCTION:

Feed the roller with concrete in the complicated areas (curved) of the tile.

The stirrer pin must avoid the creation of **vaults** when the concrete enters the roller.

It's common to adjust the stirrer pin to the tile sides, since the interlocking area usually has low thickness in the tile or in the area of the roller's smaller diameter.

I suggest to distribute the stirrer pins in such a way that they feed more the curved areas, because the flat areas are less necessary.



LIMITERS FUNCTION:

The limiter polycors must hold the ingress of concrete in certain areas to avoid feed excess which later gives way to a compaction excess.

The purpose of the limiters polycors is to try to achieve the same pressure/compaction along the entire tile cutting line.

I suggest with the new distribution of stirrers is to halve the length of the limiters, in case of excessive pressure in the flat areas, you would have to go back to the original length or longer.



ELECTROMECHANICAL CUTTER “DOUBLE ROMAN”

Attention must be paid to the sliding guides and the sliding guides of the cutter.

It is not recommended to clean the cutting carriage contour with high pressure water.

The carriage and cutting slide shafts must be kept clean and periodically lubricated.

